

(2 Hours)

[Total Marks: 60]

- N. B.: (1) All questions are compulsory.
 (2) Make suitable assumptions wherever necessary and state the assumptions made.
 (3) Answers to the same question must be written together.
 (4) Numbers to the right indicate marks.
 (5) Draw neat labeled diagrams wherever necessary.
 (6) Use of Non-programmable calculators is allowed.

1. **Attempt any two of the following:** 12
 - a. What does CAP stand for in the CAP theorem? Discuss.
 - b. Compile a list of advantages and disadvantages of using NoSQL databases.
 - c. Identify real-world examples where Big Data poses challenges.
 - d. Explain the design decisions considered for MongoDB.

2. **Attempt any two of the following:** 12
 - a. Assess the use cases where capped collections provide significant benefits.
 - b. List a set of guidelines for data import in shared environments.
 - c. What is the purpose of the `_id` field in MongoDB documents?
 - d. Discuss the structure and syntax of query documents.

3. **Attempt any two of the following:** 12
 - a. What are the advantages of using GridFS?
 - b. Explain the use of the following commands with examples:
 - i. `mongostat` ii. `Mongotop`
 - c. Discuss the following with respect to MongoDB:
 - i. Joins ii. Memory Usage iii. Document Size and testing iv. Duplicate field names
 - v. ACID compliance vi. Querying the whole database vii. Simultaneous connections
 - viii. namespace name
 - d. How does MongoDB provide durability in the event of crash? How does this affect the performance? Explain.

4. **Attempt any two of the following:** 12
 - a. What is meant by disk economics?
 - b. Discuss the benefits of using jQuery for web development.
 - c. How does jQuery simplify event management? Give the output of the following - Return the data passed with the `on()` method for each `<p>` element


```
$( "p" ).each( function( i ) {
    $( this ).on( "click", { x: i }, function( event ) {
        alert( "The " + $( this ).index() + ". paragraph has data: " + event.data.x );
    } );
} );
```
 - d. Describe a scenario where using Oracle 12C's in-memory feature would be advantageous.

5. Attempt any two of the following:**12**

- a. What are the basic data types in JSON?
 - b. What is JSONP? Where is it used? Explain in detail.
 - c. Give an overview about JavaScript Object Notation (JSON). Explain JSON tokens.
 - d. What is the use of stringify() function? Explain with example. How is it used with array?
Write a script to store Javascript objects as text using stringify().
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